NorthWest Architecture Studio Network

COM4401 Mini Project 3: Networking Fundamentals (team task)

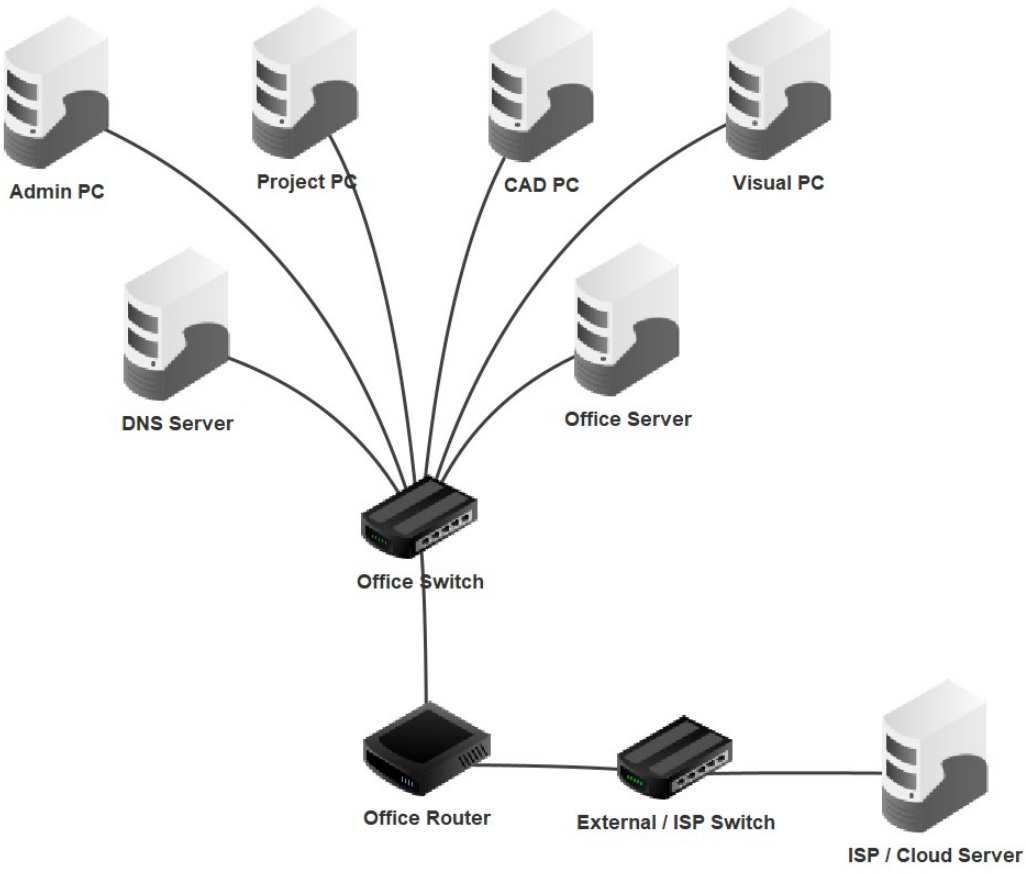
The Avengers (Team 4)

9 October 2026

1. Topology

Screenshot 1: the whole network

The office LAN (192.168.10.0/24) and the external / ISP network (203.0.113.0/24), joined by the Office Router.



Screenshot 1: the four staff PCs, the DNS Server and the Office Server on the Office Switch; the Office Router joins the Office Switch to the External / ISP Switch and the ISP / Cloud Server.

2. IP addressing table

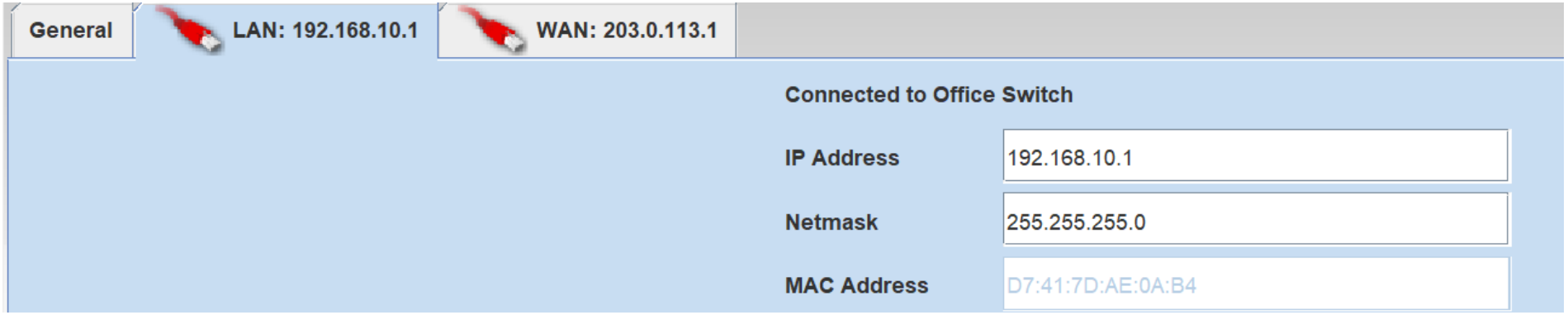
Device	IP address	Subnet mask	Gateway	DNS
Office Router (office side)	192.168.10.1	255.255.255.0	none	none
Office Router (external side)	203.0.113.1	255.255.255.0	none	none
DNS Server	192.168.10.2	255.255.255.0	192.168.10.1	192.168.10.2
Office Server	192.168.10.10	255.255.255.0	192.168.10.1	192.168.10.2
Admin PC	192.168.10.101	255.255.255.0	192.168.10.1	192.168.10.2
Project PC	192.168.10.111	255.255.255.0	192.168.10.1	192.168.10.2
CAD PC	192.168.10.121	255.255.255.0	192.168.10.1	192.168.10.2
Visual PC	192.168.10.131	255.255.255.0	192.168.10.1	192.168.10.2
ISP / Cloud Server	203.0.113.10	255.255.255.0	203.0.113.1	192.168.10.2

The two switches have no IP address.

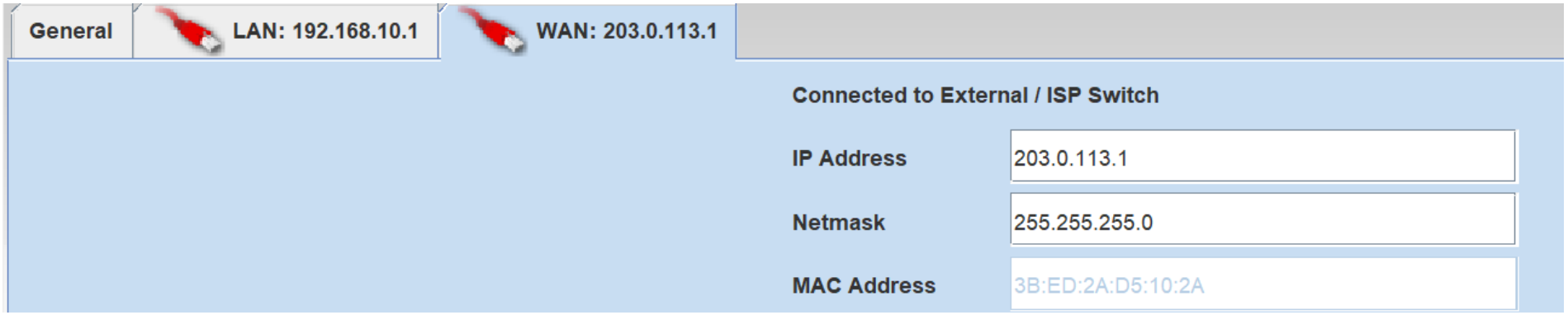
DNS records

DNS name	Points to	Meaning
intranet.archstudio.local	192.168.10.10	Internal office intranet server
cloud.archstudio.example	203.0.113.10	External cloud service server

3. Router interfaces

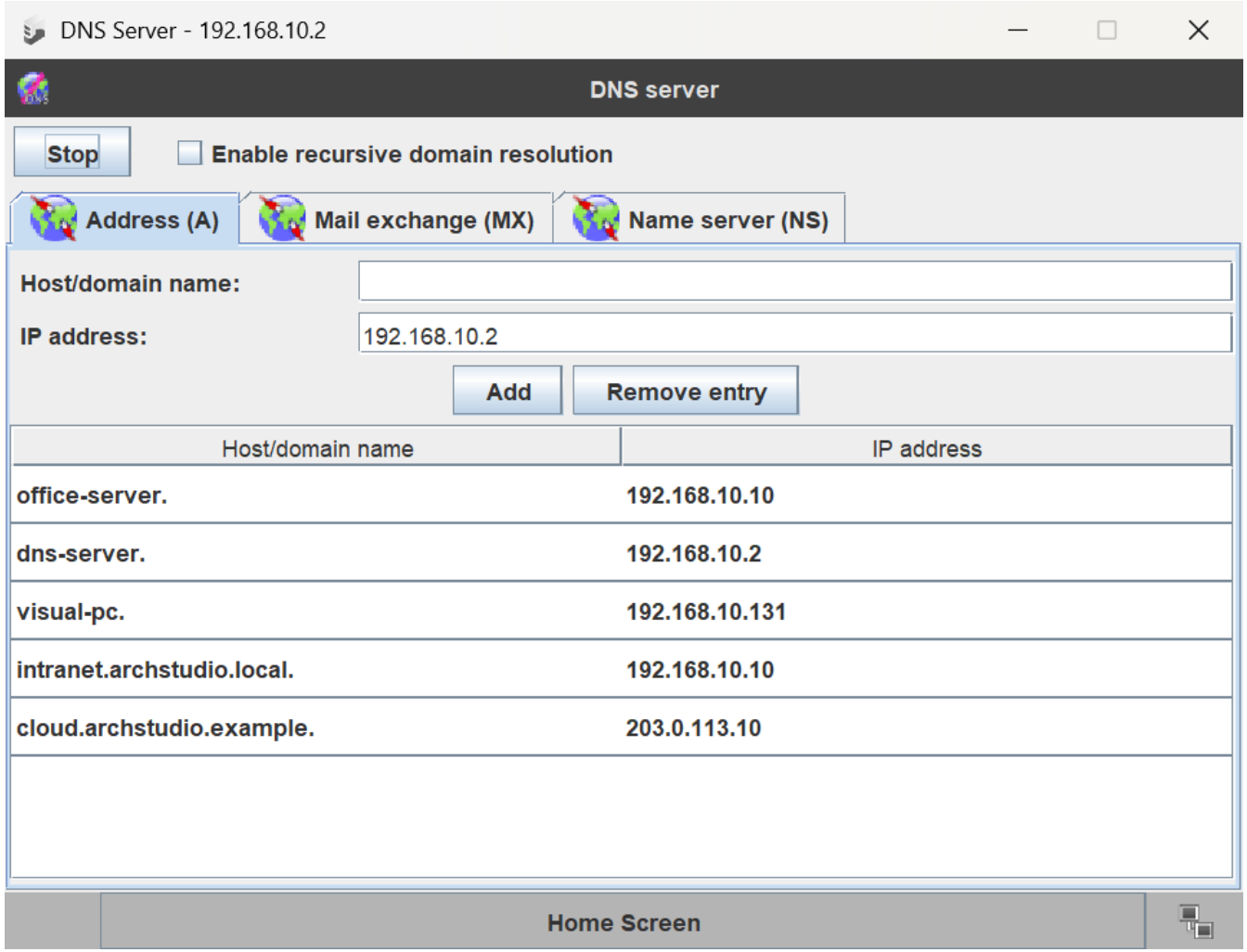


Screenshot 2: the Office Router's office side (LAN), 192.168.10.1 / 255.255.255.0, connected to the Office Switch.



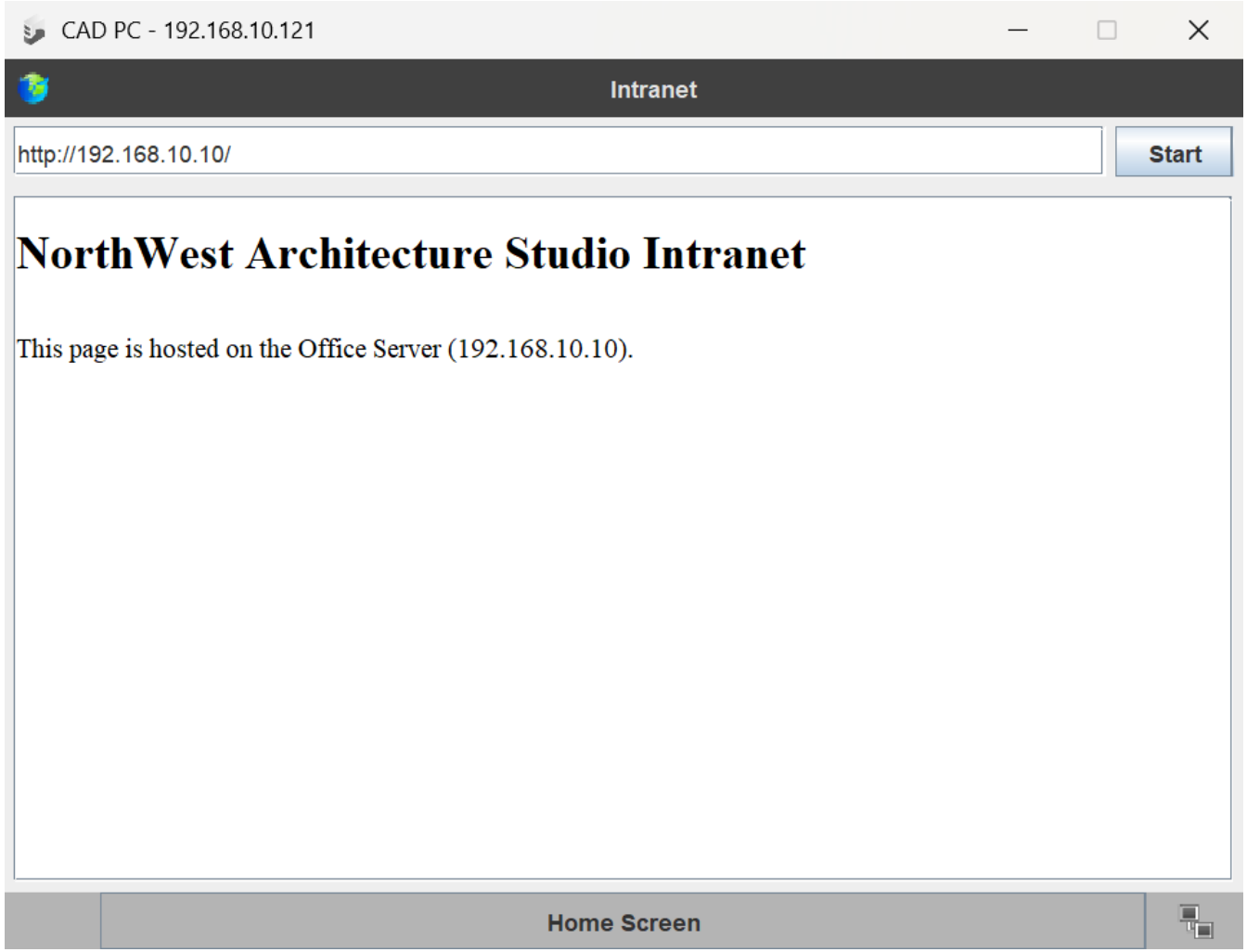
Screenshot 3: the Office Router's external side (WAN), 203.0.113.1 / 255.255.255.0, connected to the External / ISP Switch.

4. DNS records

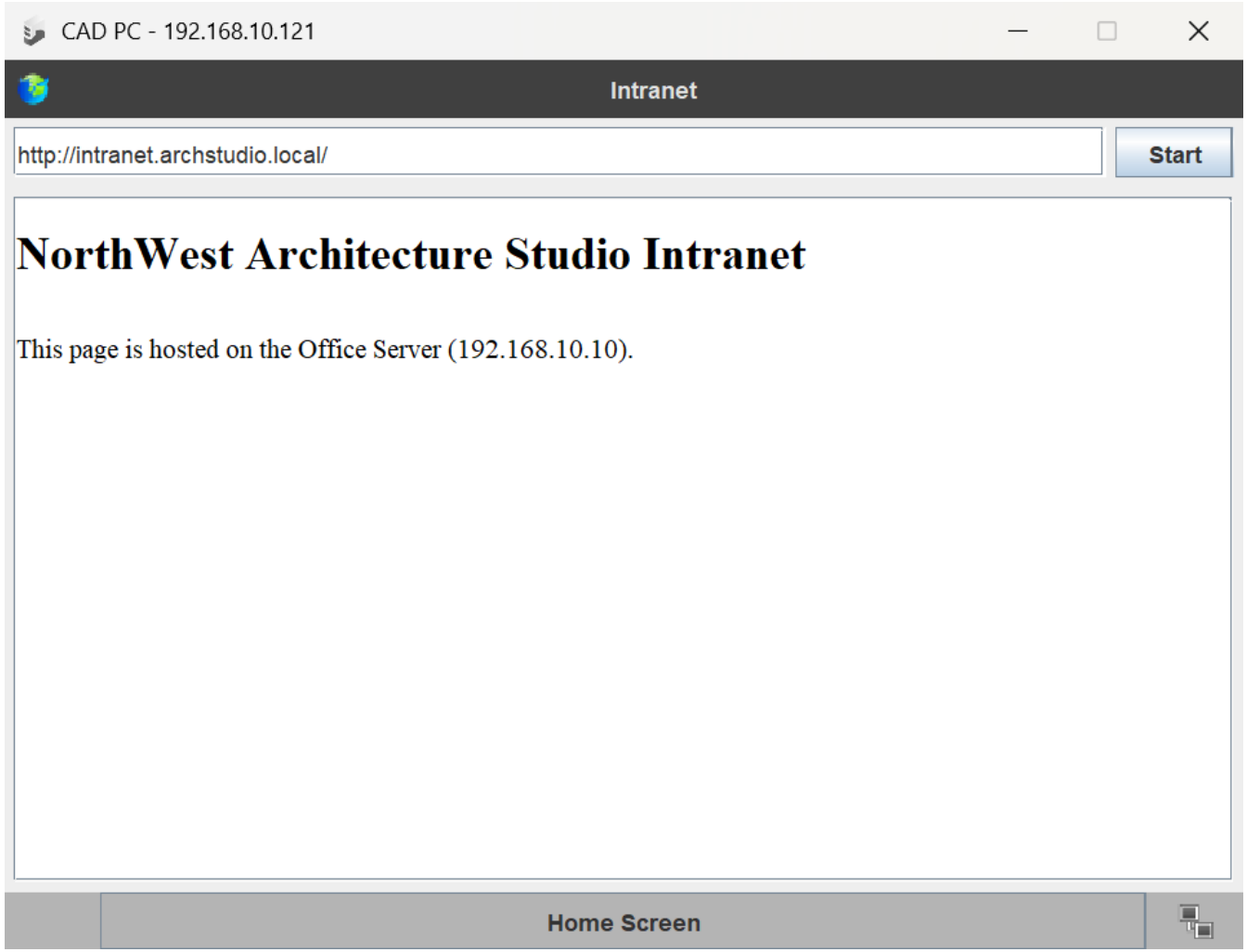


Screenshot 4: the DNS Server's address (A) records, including intranet.archstudio.local → 192.168.10.10 and cloud.archstudio.example → 203.0.113.10.

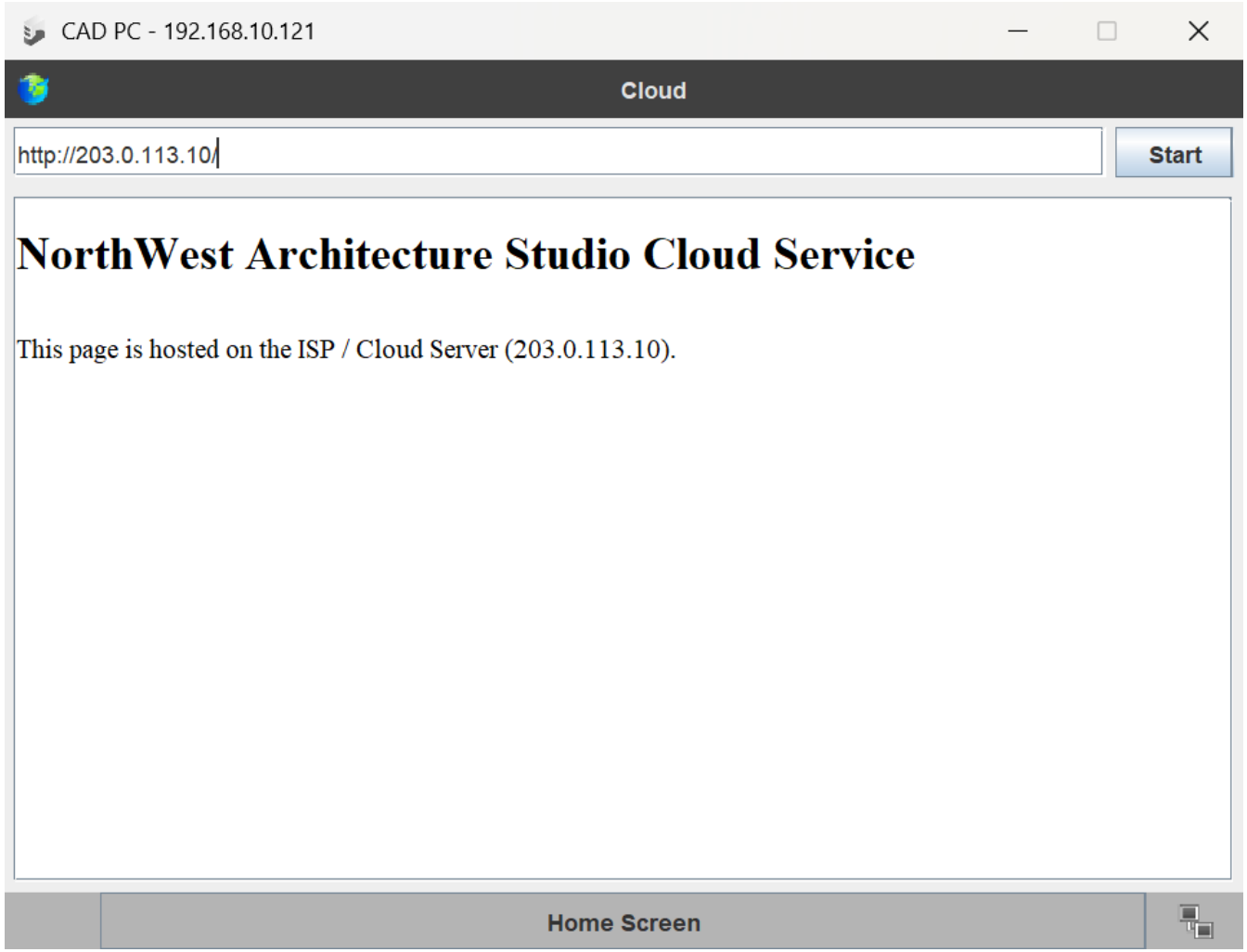
5. Web pages opening from the CAD PC



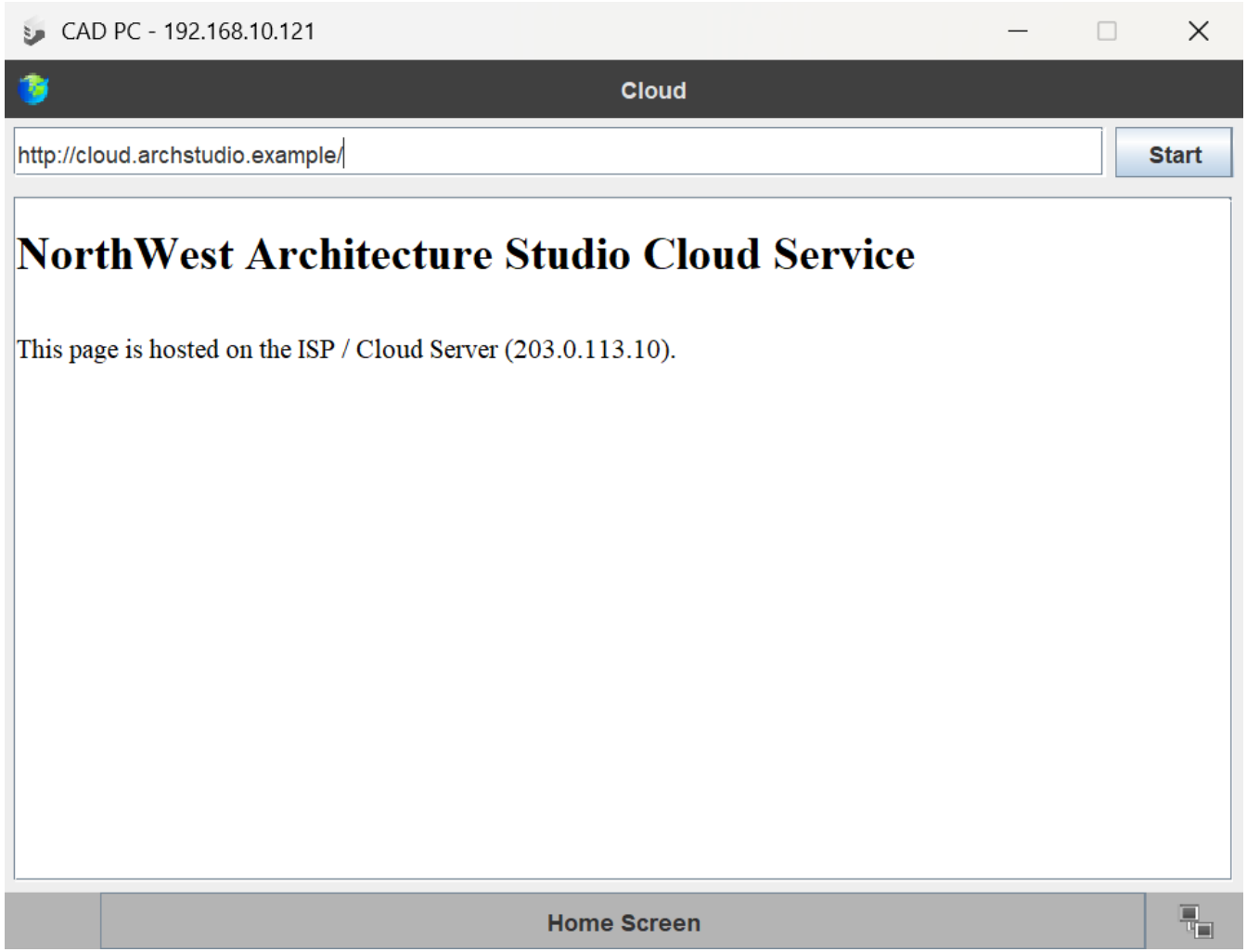
Screenshot 5 (test 4): the CAD PC's browser opening the intranet page from http://192.168.10.10.



Screenshot 6 (test 5): the same intranet page opened by name, http://intranet.archstudio.local.

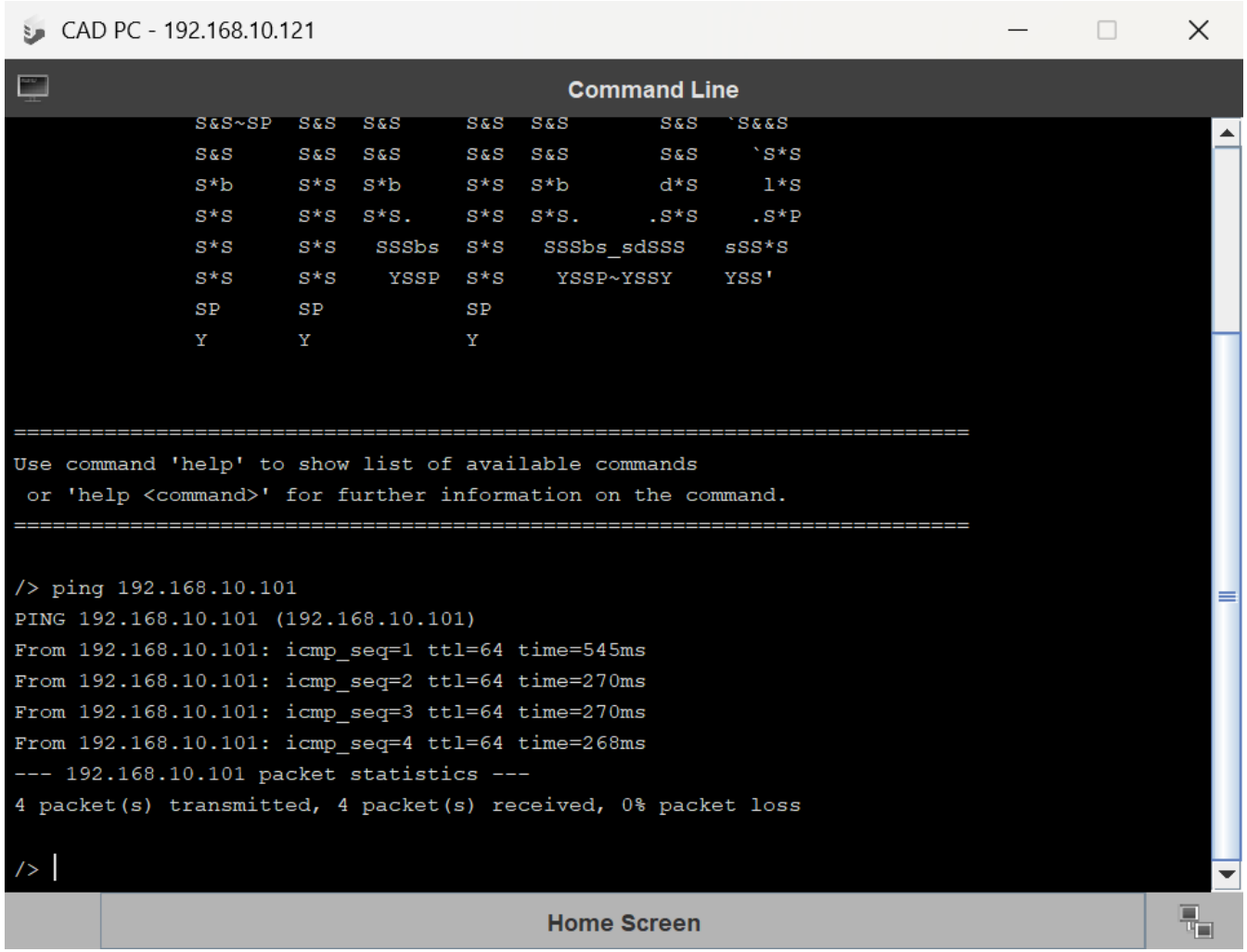


Screenshot 7 (test 7): the CAD PC's browser opening the cloud page from http://203.0.113.10, through the Office Router.

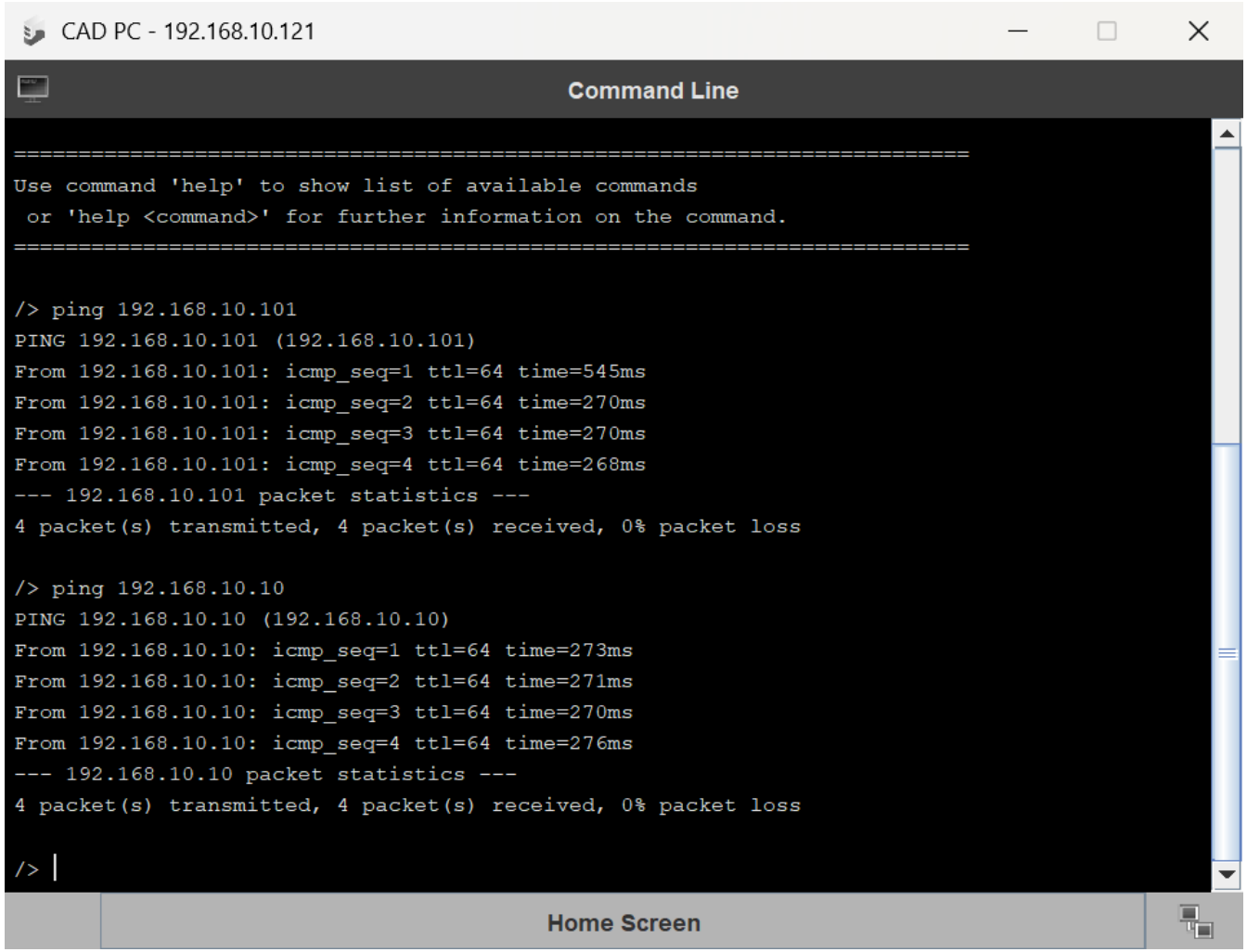


Screenshot 8 (test 8): the same cloud page opened by name, http://cloud.archstudio.example.

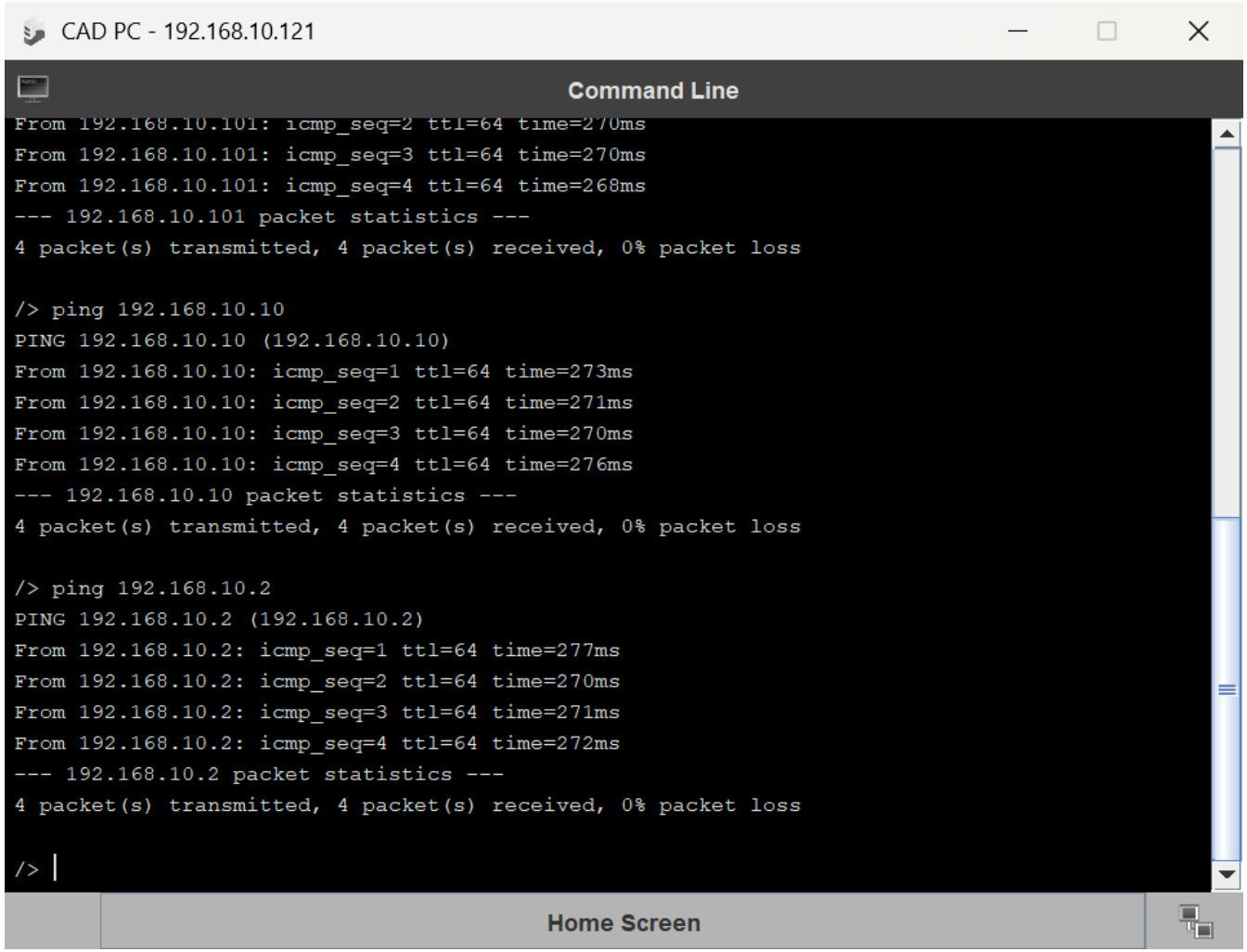
6. Ping tests from the CAD PC



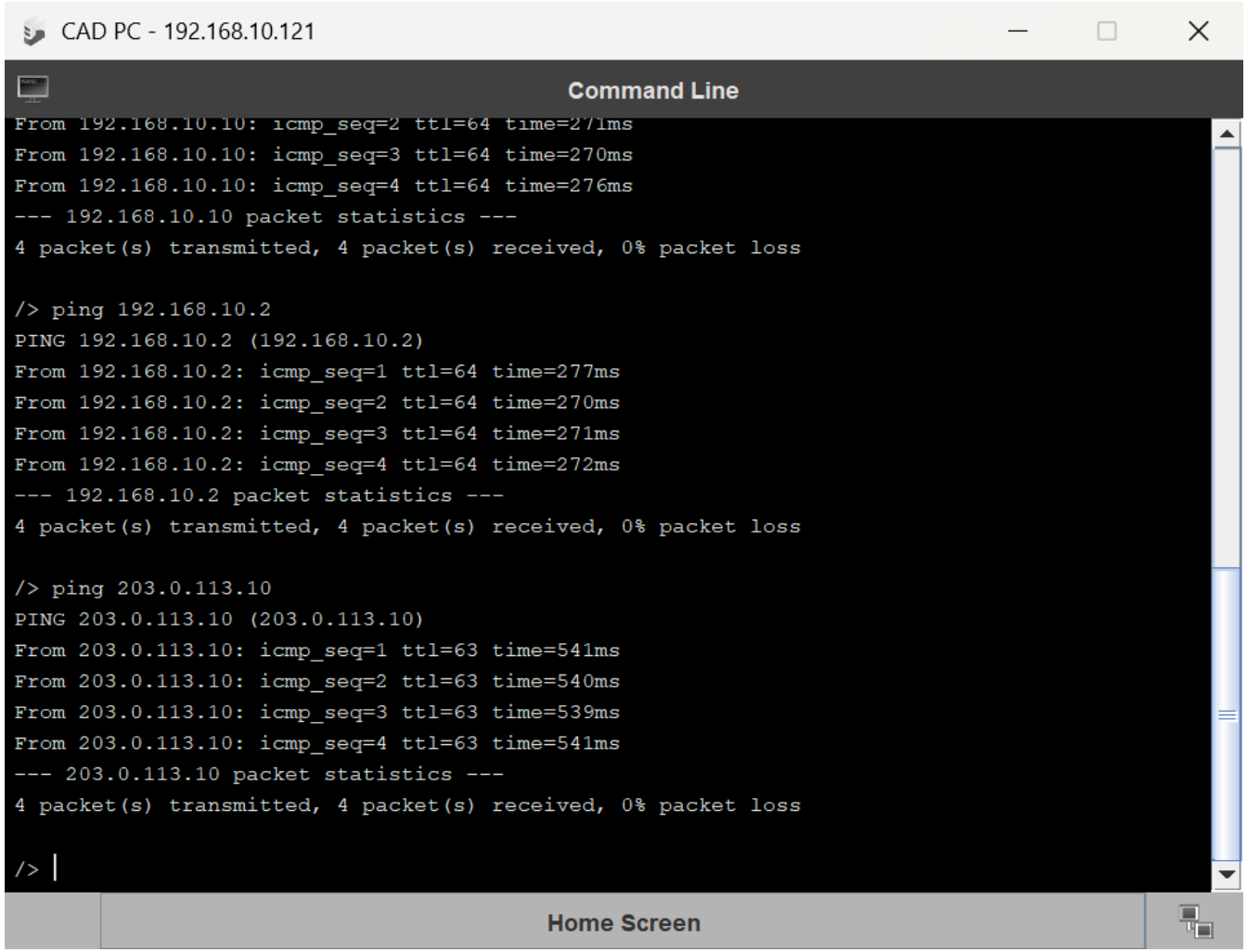
Screenshot 9 (test 1): ping 192.168.10.101 (Admin PC), 4 of 4 replies, 0% packet loss.



Screenshot 10 (test 2): ping 192.168.10.10 (Office Server), shown at the bottom of the window, 4 of 4 replies, 0% packet loss.



Screenshot 11 (test 3): ping 192.168.10.2 (DNS Server), shown at the bottom of the window, 4 of 4 replies, 0% packet loss.



Screenshot 12 (test 6): ping 203.0.113.10 (ISP / Cloud Server on the external network), shown at the bottom of the window, 4 of 4 replies, 0% packet loss.

7. Testing table

All tests were run from the CAD PC.

No.	Test	Command / action	Expected result	What it proves	Actual result	Pass / Fail
1	LAN PC to PC	ping 192.168.10.101	4 replies from 192.168.10.101, 0% packet loss	Office LAN devices communicate through the switch	4 of 4 replies from 192.168.10.101, 0% packet loss (Screenshot 9)	Pass
2	PC to Office Server	ping 192.168.10.10	4 replies from 192.168.10.10, 0% packet loss	The client can reach the internal server	4 of 4 replies from 192.168.10.10, 0% packet loss (Screenshot 10)	Pass
3	PC to DNS Server	ping 192.168.10.2	4 replies from 192.168.10.2, 0% packet loss	The client can reach the DNS service	4 of 4 replies from 192.168.10.2, 0% packet loss (Screenshot 11)	Pass
4	Internal web by IP	http://192.168.10.10	The Office Server’s index.html page opens	HTTP client-server works inside the office	Intranet page opened (Screenshot 5)	Pass
5	Internal web by name	http://intranet.archstudio.local	The same Office Server page opens	DNS turns the internal name into its IP address	Same intranet page opened by name (Screenshot 6)	Pass
6	External server ping	ping 203.0.113.10	4 replies from 203.0.113.10, 0% packet loss	The router connects the office LAN to the external network	4 of 4 replies from 203.0.113.10, 0% packet loss (Screenshot 12)	Pass
7	External web by IP	http://203.0.113.10	The ISP / Cloud Server’s index.html page opens	The client can use an external service	Cloud page opened (Screenshot 7)	Pass
8	External web by name	http://cloud.archstudio.example	The same ISP / Cloud Server page opens	DNS turns the external service name into its IP address	Same cloud page opened by name (Screenshot 8)	Pass

8. Protocols and concepts

Concept	Where it appears in our network	What it does
IP addressing	Every device has its own address, for example the CAD PC is 192.168.10.121 and the ISP / Cloud Server is 203.0.113.10	IP carries the sender and destination address in each packet, so each packet can be sent toward the right place (Postel, 1981b).
Subnet mask	255.255.255.0 on both networks	A computer uses the mask to compare the destination address with its own. If the network parts match, it sends the packet straight to that device. If they do not match, it sends the packet to the gateway (Mogul and Postel, 1985).
Gateway	The office devices use 192.168.10.1 and the ISP / Cloud Server uses 203.0.113.1	The gateway is where a device sends packets bound for another network (Mogul and Postel, 1985). In our network the gateway is the Office Router.
DNS	The DNS Server (192.168.10.2) holds intranet.archstudio.local and cloud.archstudio.example	DNS turns a host name into an IP address, and the name server answers each question from the records it holds (Mockapetris, 1987). Tests 5 and 8 opened both pages by name.
HTTP	The CAD PC’s web browser opening the two web pages (tests 4, 5, 7 and 8)	The browser sends an HTTP request and the web server replies with an HTTP response that carries the page (MDN Web Docs, 2026).
ICMP / ping	Tests 1, 2, 3 and 6 from the CAD PC	Ping sends an ICMP echo message, and the other device has to send the same data back in an echo reply (Postel, 1981a). Every ping got 4 of 4 replies.

9. LAN vs external network, and the client-server model

The office LAN is 192.168.10.0/24. It holds the four staff PCs, the DNS Server and the Office Server, all plugged into the Office Switch. A LAN is a set of devices connected together in one place, such as an office (Cisco, N.D.a). Addresses starting 192.168 are private, so any company can use them inside its own network, and they are only unique inside that company (Rekhter et al., 1996).

The outside network is 203.0.113.0/24. It holds only the ISP / Cloud Server on the External / ISP Switch, and it stands in for the internet. That block of addresses is kept for examples and should not appear on the real internet (Arkko et al., 2010), which makes it a safe choice for a simulation.

The Office Router joins the two networks. It has an address on each side: 192.168.10.1 for the office and 203.0.113.1 for the outside. Every office device uses 192.168.10.1 as its gateway. The router reads the destination IP address in each packet and chooses the route it takes (Cisco, N.D.b).

In the web page tests, the CAD PC is the client. It runs the web browser and asks for the page. The Office Server and the ISP / Cloud Server are the servers, because they run the web server software, wait for requests and send pages back. The browser sends an HTTP request and the server replies with an HTTP response that carries the page (MDN Web Docs, 2026).

References

Arkko, J., Cotton, M. and Vegoda, L. (2010) *IPv4 address blocks reserved for documentation.* [Online] RFC Editor. Available from: https://www.rfc-editor.org/rfc/rfc5737.html. [Accessed 9 October 2026].

Cisco. (N.D.a) *What is a LAN? Local area network.* [Online] Available from: https://www.cisco.com/site/us/en/learn/topics/networking/what-is-a-lan-local-area-network.html. [Accessed 9 October 2026].

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